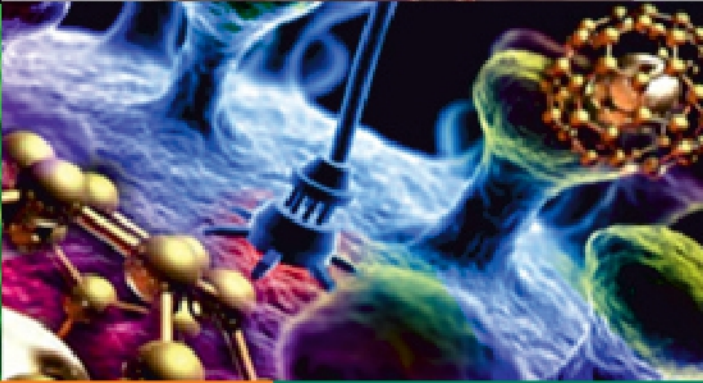
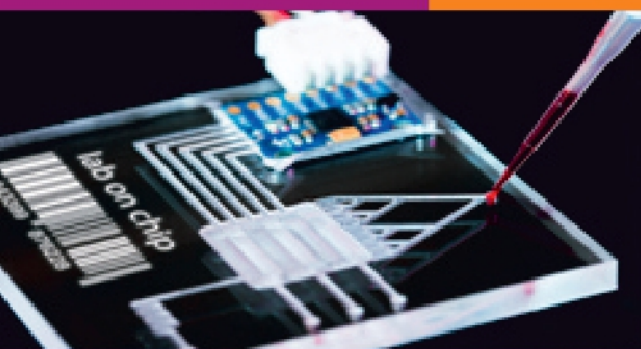


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ABSTRACTS



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Development of patterned paper devices with specified surface characteristics for bioanalytical applications.

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Abstract: This paper introduces an innovative technique for creating hydrophobic patterns on paper surfaces, enabling the development of paper-based analytical devices for point-of-care diagnostics. The sensor's reaction zone was modified to be hydrophilic while surrounding areas were strategically made hydrophobic using an agarose layer and subsequent polydimethylsiloxane (PDMS) deposition. Circular reaction zones undergo layer-by-layer agarose deposition, effectively masking the surface against PDMS layering. Scanning Electron Microscopy (SEM) analysis demonstrates successful capillary blocking in PDMS-rendered hydrophobic paper areas. Subsequently, an assay reagent specific to glucose detection is loaded into the reaction zones, leading to a distinct color change from colorless to blue through the glucose oxidase-catalyzed reaction, indicating the presence of glucose. Incorporating SiO₂ NPs in the assay formulation further enhances the sensing response. This novel approach shows enormous potential for creating patterned paper platforms in point-of-care diagnostics, particularly in resource-limited settings.

Keywords: Paper-based analytical devices, hydrophobic patterns, agarose, polydimethylsiloxane (PDMS), Bioanalytical applications.

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